

DISCRETE MATHEMATICS

FOR
BS 4-YEARS



Z.R. BHATTI



Chapter-4 **RELATIONS**

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(v) t : Mobile phone A has a 10GB memory card.

(vi) u : His motorcycle travels 60 km in one litre of petrol.

Solution: The negations of above propositions are given below.

(i) $\neg p$: 121 is not a perfect square.

(ii) $\neg q$: $2 + 7 = 9$

(iii) $\neg r$: $2 + 7 \neq 9$

(iv) $\neg s$: Karachi is not a capital of Sindh.

(v) $\neg t$: Mobile phone A has not 10GB memory card.

(vi) $\neg u$: His motorcycle does not travel 60 km in one litre of petrol.

Truth Table for Negation of a Proposition:

While constructing the truth table for negation of a proposition, we write the proposition p and its truth values in the first column and the negation of p , $\neg p$, and its truth values in the second column. In the following, truth table for negation of a proposition is given.

p	$\neg p$
T	F
F	T

Truth Table for the Negation of a Proposition

Example 3: Construct the truth table for the negations of the following propositions.

(i) p : 79 is a perfect square.

(ii) q : $2 + 7 \neq 9$

(iii) r : $2 + 7 = 9$

(iv) s : Karachi is a capital of Sindh.

(v) t : $3 > 5$

(vi) u : New York is a state of USA.

Solution: Since

p is false, so $\neg p$ is true;

q is false, so $\neg q$ is true;

r is true, so $\neg r$ is false;

s is true, so $\neg s$ is false;

t is false, so $\neg t$ is true;

u is true, so $\neg u$ is false.

The truth table is given below.

p	$\neg p$	q	$\neg q$	r	$\neg r$	s	$\neg s$	t	$\neg t$	u	$\neg u$
F	T	F	T	T	F	T	F	F	T	T	F

The negation operator, $\neg$, constructs a new proposition from a single existing proposition. In the following, the logical operators are introduced that are used to form new propositions from two or more existing propositions. These logical operators are also called *connectives*.

Conjunction of Propositions: A compound proposition of two propositions p and q is called *conjunction* of p and q if it is true only if both p and q are true and is false otherwise. Conjunction of p and q is read as

" p and q ."

It is denoted by $p \wedge q$.

In other words, the conjunction of p and q is denoted by $p \wedge q$ and is true only if both p and q are true and is false otherwise.

Following is the truth table for conjunction of two propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 4: Construct the truth table for $p \wedge q$ where
 p : 11 is divisible by 2.

q : Quaid-i-Azam was founder of Pakistan.

Solution: Since 11 is not divisible by 2, so p is false.
 q is true.

The truth table for $p \wedge q$ is given below.

p	q	$p \wedge q$
F	T	F

Example 5: Find the conjunction of the propositions p and q where

p : Azeem's PC has more than 40 GB free hard disk space.

q : The processor in Azeem's PC runs faster than 1 GHz.

Solution: The conjunction of these propositions, is the proposition

$p \wedge q$: Azeem's PC has more than 40 GB free hard disk space, and the processor in Azeem's PC runs faster than 1 GHz.

This conjunction can be expressed more simply as

$p \wedge q$: Azeem's PC has more than 40 GB free hard disk space, and its processor runs faster than 1 GHz.

For this conjunction to be true, both conditions given must be true. It is false, when one or both of these conditions are false.

Disjunction of Propositions: A compound proposition of two propositions p and q is called *disjunction* of p and q if it is false only if both p and q are false and is true otherwise. Disjunction of p and q is read as

" p or q ."

It is denoted by $p \vee q$.

In other words, the disjunction of p and q is denoted by $p \vee q$ and is false only if both p and q are false and is true otherwise.

Following is the truth table for disjunction of two propositions.

















































































































































